

Lesson 3: Vector Data with shapely and geopandas

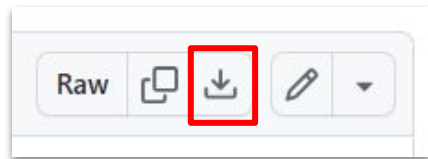
As you arrive and get set up...



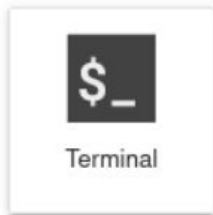
1. Open **CryoCloud**: hub.cryointhecloud.com
 - Start a 1.9GB Python node
 - Create a folder named `lessons` (or any way you want to organize it)

2. Make a copy of **today's lesson notebook**:

- In your web browser, visit:
https://github.com/NASA-SARP/2026-Coding-Alex/blob/main/lessons/03_vector_data.zip
- Download the zip folder `03_vector_data.zip` and unzip:
- “Drag and drop” the unzipped folder `03_vector_data` into the `lessons` folder you created earlier in CryoCloud



[2. *Advanced option*]*—*use the terminal to copy the folder in CryoCloud instead:



```
cp -ri ~/shared-public/NASA-SARP/2026-Coding-Alex/lessons/03_vector data  
~/Projects/lessons
```

← → ↺

github.com/NASA-SARP/2026-Coding-Alex/blob/main/lessons/03_vector_data.zip

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📄 01_building_blocks.zip

2026-Coding-Alex / lessons / 03_vector_data.zip 📄

 alex-saunders00 Create 03_vector_data.zip e2cf9a7 · 14 hours ago 🕒 History

Code Blame

Raw 📄 ⬇️ ✎ ▾

View raw

(Sorry about that, but we can't show files that are this big right now.)

Recap: Goals for our coding lessons

Data types



Data structures



Keeping track of individual variables (pieces of data) would get very out of hand very quickly → to organize our data in coding we build **data structures**

Recap: From data types to data structures

Data types

Data type	Python name	What it holds	Examples
Integer	int	Whole numbers	14, 0, -3
Float	float	Decimal numbers	3.5, -0.2, 6.022e23
String	str	Text	"Jack", 'hello'
Boolean	bool	True or False	True, False

Data structures

List

```
mixed = ['abc', 34, True, 40, 'male']
```

Dictionary

```
metadata = {  
    'location': 'Nairobi',  
    'quality_flag': 5,  
    'zenith': 60,  
    'clouds': True,  
}
```

DataFrame

	temperature	humidity	wind_speed
0	25	60	10
1	20	65	9
2	27	62	10
3	31	80	6

GeoDataFrame

	COUNTYFP	NAME	geometry
0	001	Alameda	POLYGON ((-13612246.763 4538150.085, -13612346...
1	003	Alpine	POLYGON ((-13366503.508 4678946.015, -13366492...
2	005	Amador	POLYGON ((-13472697.863 4647652.232, -13472697...
3	007	Butte	POLYGON ((-13565004.793 4798394.124, -13564991...
4	009	Calaveras	POLYGON ((-13428575.483 4627725.227, -13428534...

Recap: Anatomy of a `pd.DataFrame`

`df =`

	temperature	humidity	wind_speed
0	25	60	10
1	20	65	9
2	27	62	10
3	31	80	6

← Column
names

← A single
row

↑
Indexes

↑
A single column—a `pd.Series`

Adding spatial information → `gpd.GeoDataFrame`

`[]` indexing

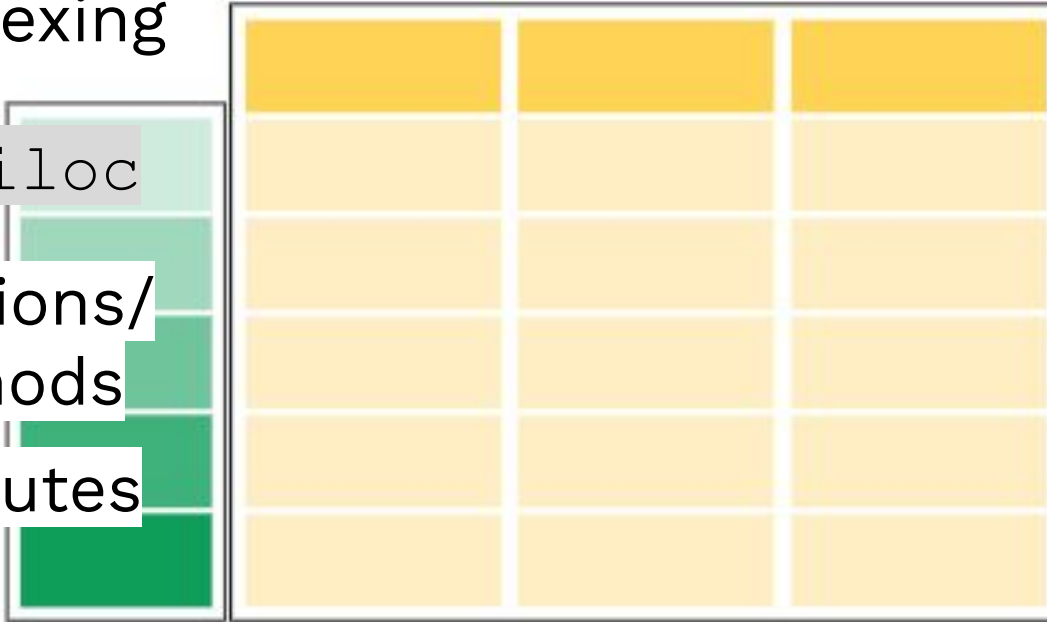
`loc, iloc`

Functions/
methods

Attributes

index

data

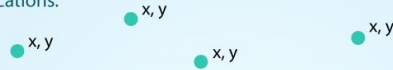


Types of spatial data

Vector geometries

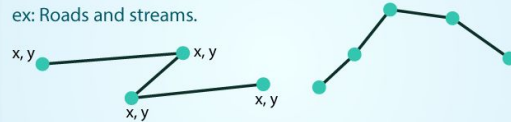
POINTS: Individual x, y locations.

ex: Center point of plot locations, tower locations, sampling locations.



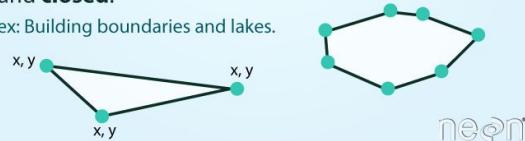
LINES: Composed of many (at least 2) vertices, or points, that are connected.

ex: Roads and streams.

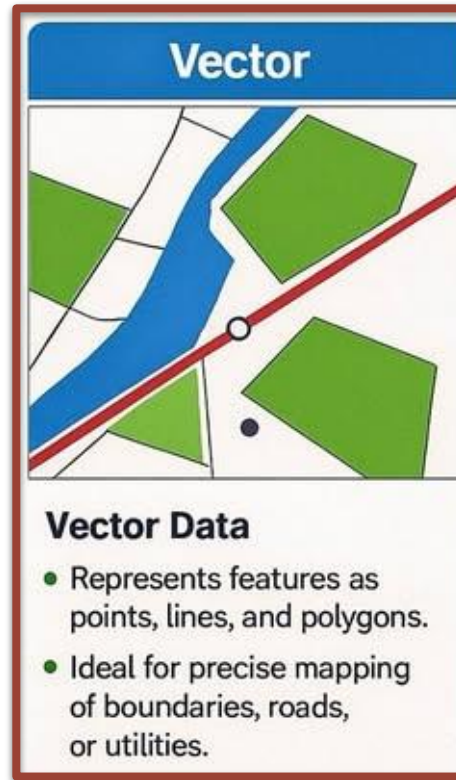


POLYGONS: 3 or more vertices that are connected and **closed**.

ex: Building boundaries and lakes.



neon

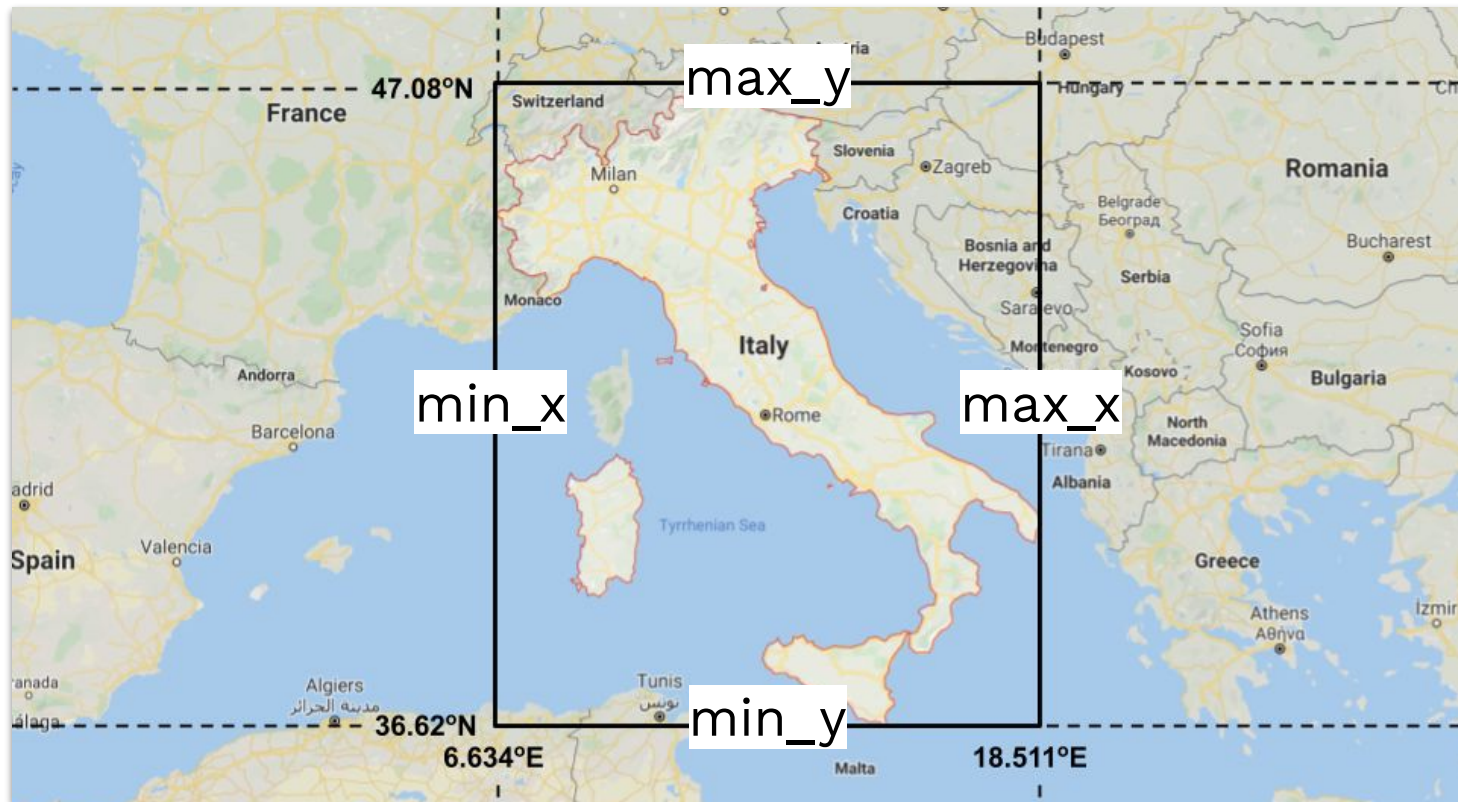


```
from shapely import Point, LineString, Polygon, box
```

library

geometry classes

Special mention: bounding box



`gpd.GeoDataFrame`



GeoPandas

```
import geopandas as gpd
```

`gdf =`

	COUNTYFP	NAME	geometry
0	001	Alameda	POLYGON ((-13612246.763 4538150.085, -13612346...
1	003	Alpine	POLYGON ((-13366503.508 4678946.015, -13366492...
2	005	Amador	POLYGON ((-13472697.863 4647652.232, -13472697...
3	007	Butte	POLYGON ((-13565004.793 4798394.124, -13564991...
4	009	Calaveras	POLYGON ((-13428575.483 4627725.227, -13428534...

GeoDataFrame = [ +   =    f shapely objects

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Please complete the **exit ticket**:

<https://forms.gle/UuwpwSEoqDbV5oGo9>

Don't forget to **shut down CryoCloud**:

File → Hub Control Panel → Stop Server